

Title/Citation (ACM auto generated)	Goal	Key Findings	Future	Gaps/Notes
[1] Sarah Tabassum, Narges Zare, and Cori Faklaris. From Scam to Safety: Participatory Design of Digital Privacy and Security Tools with International Students from Global South.	Investigate digital challenges and find desired solutions of educational migrants from Global south	Participants wanted reporting options, verification of threats, university tools and educations to help newly assimilated educated migrants	Design safety interventions that move throughout the digital ecosystem.	
[1] Cori Faklaris. 2024. Mitigating Smishing: Challenges and Future Work. https://doi.org/10.48550/arXiv.2401.14520	Identify challenges in smishing	Device limitations, complexity of network infrastructure, context/cognitive factors	Develop new interfaces, warning systems, education, and reporting	
[1] Seyed Mohammad Sanjari, Ashfak Md Shibli, Maraz Mia, Maanak Gupta, and Mir Mehedi Ahsan Pritom. 2025. SmishViz: Towards A Graph-based Visualization System for Monitoring and Characterizing Ongoing Smishing Threats. In Proceedings of the Fifteenth ACM Conference on Data and Application Security and Privacy (CODASPY '25), June 04, 2025. Association for Computing Machinery, New York, NY, USA, 257–268. https://doi.org/10.1145/3714393.3726499	To uncover and collect data on smishing campaigns and their operations by collecting data using visualization and data mining techniques (clustering) to understand the threat ecosystem.	Smishing campaigns are large scale, coordinated operations, not just simple isolated scams. Common theme of reuse between attacks, identical domains, impersonations, and use of URL shorteners 88.6% contained URLs, 65.9% impersonated brands, 182 unique brands were used	SmishViz has plans to continuously build and collect data on smishing. Scalability, new clustering, more graphs and analysis.	
[1] Md Lutfor Rahman, Daniel Timko, Hamid Wali, and Ajaya Neupane. 2023. Users Really Do Respond To Smishing. In Proceedings of the Thirteenth ACM Conference on Data and Application Security and Privacy (CODASPY '23), April 24, 2023. Association for Computing Machinery, New York, NY, USA, 49–60. https://doi.org/10.1145/3577923.3583640	Investigate why smishing works, what smishing messages works better, what demographics are affected more.	Smishing is effective! 16.92% responded first round, 12.82% responded second round certain messages increase success rate (facebook + walmart messages) 18-24 yer olds has lower odds of falling for smishing african and mexican americans had the highest odds Participants interacted even if they knew it was a scam	Better detection systems are needed, more education is needed to teach people how to interact with smishing	
[1] Sarah Tabassum, Cori Faklaris, and Heather Richter Lipford. What Drives SMiShing Susceptibility? A U.S. Interview Study of How and Why Mobile Phone Users Judge Text Messages to be Real or Fake.	Understand how users judge SMS messages to be real or fake, what factors into smishing judging (cybertraining, message-context)	Users look at links, messages, area codes to judge common scams include package delivery, financial fraud, business promotion, impersonation Legitimacy includes personalized info, known context, known sender, official formats visual warnings and cybertraining helps! lack of reporting	Need for filters and warnings. Reporting options need to be easier More research needed Cybertraining as defense helps	
[1] Ezer Osei Yeboah-Boateng and Priscilla Mateko Amanor. 2014. Phishing, SMiShing & Vishing: An Assessment of Threats against Mobile Devices. 5, 4 (2014).	Conduct an assessment of new social engineering attacks (smishing, vishing, and phishing) targetting mobile devices and figure out how many of these attacks happen, why, and how people fall victim to these attacks	Men were more susceptible to phishing attacks overall. Most people have never heard of smishing and 35% of participants admitted they don't even scrutinize any messages whatsoever. iOS is found to be the most susceptible to these attacks. Common attack words "click here, (links), download, account, password, document, transaction	SETA (security education, training and awareness) would help reduce victim numbers. Emphasis on future work needs to be focused on behavior interactions rather than just technical filters	This is an early study that was conducted in 2014 when these new types of attacks were becoming more prevalent. The study only had 20 participants The study has participants only from Ghana.

[1] David Ng'ang'a Njuguna, John Kamau, and Dennis Kaburu. 2022. A Review of Smishing Attacks Mitigation Strategies. IJCIT 11, 1 (March 2022). https://doi.org/10.24203/ijcit.v11i1.201	Conduct a review of security challenges facing mobile devices as well as what threats mobile devices face through smishing. Conduct a review of the possible solutions and what technological solutions have already been proposed and worked on.	SMS messages have a high response rate compared to email. Most cyber attacks are due to human error Best mitigation techniques: Education and Technology solutions	Enhance user awareness and education, create hybrid solutions that include both education and tech solutions	No single technique is capable of mitigating 100% of smishing techniques
[1] Elif Ustundag Soykan and Mustafa Bagriyanik. 2020. The Effect of SMiShing Attack on Security of Demand Response Programs. Energies 13, 17 (September 2020), 4542. https://doi.org/10.3390/en13174542	Investigate and model a new type of cyber-physical threat called "Distrubing DR via Smishing" (DDRS) attacks. Focuses on how social engineering and smishing can be used to manipulate DR systems which can cause physical instability or power outages in a grid system without needing to breach the grid's infrastructure	Powergrids are vulnerable by Smishing Local outages were created through simulation of IEEE European Low Voltage Feeder Test System Smishing is sucessful and dangerous compared to phishing because mobile users have few tools to verify Easy to do smishing --> easy to conduct a DDRS attack	Larger scale research is needed to understand the DDRS attack. Future work needs to focus on utility companies being able to interact with customers to prevent and teach about smishing.	Definition Demand Response (DR): is a way to manage consumer demand that allows balance to persist on the grid Not much research is out there on the DDRS attack topic
[1] Bello Bello Musa, Adamu Ahmad Bahago, Nasir Auwal Muhammad, and Dr Fakhrun Jamal. 2025. Emerging Trends in Phishing: A Look at Smishing, Vishing, Quishing. 01, 01 (2025).	Study the evolution of phishing strategies beyond email ... focus on smishing, vishing, and quishing. Major focus on TTPs, tools, and psychology tricks used by attackers	All 3 attack styles are shifting to mobile platforms as phones are ubiquitous, has small screen (hides details), and high level of trust associated with phones Attackers exploit human trust and use manipulation tactics. AI threats are growing as attackers are using deepfake voices and using them to create messages Caller ID and SMS protocol lack real authentication making spoofing highly effective	Cross-channel campaigns are predicted to increase (use of smishing to win trust before using vishing) AI powered defense mechanisms using natural language processing can be created to help prevent vishing attacks	Many organizations and persons do not understand the phishing overall which leads to catastophic results and lack of reporting.
[1] Aaron Zimba, Katongo Phiri, Chimanga Kashale, and Mwiza Phiri. 2025. Unveiling deception: a socio-economic analysis of smishing attacks on mobile money transaction users. Humanit Soc Sci Commun 12, 1 (December 2025), 1880. https://doi.org/10.1057/s41599-025-06141-8	Understand the socio-economic impact of smishing attacks within Zambia assess threat social engineering tactics used by the attackers in english and bemba language. understand prior knowledge of participants, understand the threat motives, and what can be done to mitigate smishing	Age, gender, location, mobile network companies do not play a significant factor in receiving smishing messages 93% of persons have received smishing messages The primary threat motive is financial gain Urgency and politness is commonly used in threat messages	Future reseach needs to idntify which factors beyond demographics influence a users vulnerablitiy to smishing intervention strategies and education need to be employed.	Data collected in Zambia (377 participants)
[1] Ryu Saeki, Leo Kitayama, Jun Koga, Makoto Shimizu, and Kazumasa Oida. 2022. Smishing Strategy Dynamics and Evolving Botnet Activities in Japan. IEEE Access 10, (2022), 114869–114884. https://doi.org/10.1109/ACCESS.2022.3217795	Investigate attack strategies and patterns of two major smishing botnets targeting Japan known as XLoader and FakeSpy. Leveraging these patterns develop lightweight malware and malicious domain detection methods	Using TCP/IP flow helps but so does decoding encrypted botnet traffic using Frida hooking. Bots send state updates (sleep, silent, active) 180 sendSMS commands are devliered daily per bot at the same time traffic behavior is highly structured and routined (predictable).	Mass sms sending at the same time helps us figure out when SMshing is sent, we can use this to create malware detection. Historical patters and statistical analysis can be used a prediction to fight against smishing.	Data collected in Japan and focused on XLoader, FakeSpy

[1] Jae Woong Joo, Seo Yeon Moon, Saurabh Singh, and Jong Hyuk Park. 2017. S-Detector: an enhanced security model for detecting Smishing attack for mobile computing. Telecommun Syst 66, 1 (September 2017), 29–38. https://doi.org/10.1007/s11235-016-0269-9	Develop new security model to detect smishing that is known as the S-Detector. The purpose is to distinguish between normal text messages and smishing attempts.	S detector uses Naive Bayes Classifier to analyse frequency of nouns used in smishing messages to help determine if incoming messages are smishing attempts or not. Common blacklists are failing because threat actors typically change links frequently and shortens them Using Naive Bayes was chosen because it was faster than discriminative models like logistic regression, and required less training to maintain high accuracy.	Authors anticipate that malicious threats will be more diverse and intelligent requiring more research.	
[1] Commercial Anti-Smishing Tools and Their Comparative Effectiveness Against Modern Threats. https://doi.org/10.1145/3558482.3590173	Study effectiveness of anti-smishing tools against newly sent attacks. The purpose was to measure the performance of bulk messaging services, mobile carriers, and anti-smishing apps in how effective they are in filtering smishing.	All segments (bulk messaging services (ESME), mobile carriers, and anti-smishing apps all have gaps and need major work. Mobile carriers reached 25-35% blocking rate (T-mobile at 35%), Verizon failed to catch any attacks at all Apps like robokiller blocked smishing but also a lot of legitimate messages as well Bulk messaging services did not block messages before they were sent despite claims of following legal guidelines or helping with the smishing issue. Malicious links are often not flagged by most security vendors	Bulk messaging services need to implement stricter API approval processes to help verify legitimate business. Develops need to move away from a catch-all approach to reduce false positives	Impossible to test every bulk service, carriers, and apps
[1] Gunikhan Sonowal and K S Kuppusamy. 2018. SmiDCA: An Anti-Smishing Model with Machine Learning Approach. Comput J 61, 8 (August 2018), 1143–1157. https://doi.org/10.1093/comjnl/bxy039	Develop machine learning model that detects smishing at the message level rather than website level. Uses natural language processing.	High detection accuracy 96.40% 53.68% of smishing messages had a URL 11.57% contained an email address	Needs to continue working on this project and attempt to deploy product.	
[1] Smishing Dataset I: Phishing SMS Dataset from Smishtank.com. https://doi.org/10.1145/3626232.3653282	Provide new information and information updates of fresh smishing data to help researchers and engineers create new defenses against this ever evolving threat	Sender diversity, 23.9% of smishing was sent by email to phone number common impersonated brands are USPS, Amazon, financial institutions, netflix and contained messages about account changes, packages, advertisements, and more Typically had deceptive URLs that utilized random domains, some used "combosquatting" or "typosquatting" to appear authentic Out of 70+ security vendors, many smishing links were undetected by the current security software	Datasets are growing on smishtank.com	data set was 1062 websites are frequently taken down quickly and replaced
[1] 2025. The Anatomy of a Smishing Attack. Insights2Techinfo. Retrieved February 9, 2026 from https://insights2techinfo.com/the-anatomy-of-a-smishing-attack-common-techniques-and-tactics-used-by-cybercriminals/	Explain the structure and TTPs threat actors use in smishing attacks to determine how they operate. Establish the lifecycle of attacks	Smishing is a highly structured and systematic attack that is refined and used against victims to collect information Smishing heavily relies on social engineering to work by attacking human flaws. Filtering methods include Naive Bayesian classifiers, SMS spam filtering apps (SMS assassin), context based detection (URL detection, keywords, etc.)	A strategy has been proposed for fighting against smishing Security measures and education on Smishing, Strong MFA, legal help (creating laws and policy to prevent smishing)	

[1] W. L. T. T. N. Kumarasiri, M. K. J. C. Siriwardhana, S. A. D. S. L. Suraweera, A. N. Senarathne, and S. M. B. Harshanath. 2023. Cybersmish: A Proactive Approach for Smishing Detection and Prevention using Machine Learning. In 2023 7th International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), October 2023. 210–217. https://doi.org/10.1109/I-SMAC58438.2023.10290228	Develop machine learning solution named Cybersmish that detects and prevents smishing attacks on mobile devices. Move beyond simple message content and also evaluate the legitimacy of URLs and HTML content in the links provided from Smishing messages Enhance detection capabilities for zero day attacks using machine learning and NLP based models	MobileBERT model has a testing accuracy rate of 94.81% for mobile detection DistilBERT model has 99.82% accuracy in detecting URL danger TF-BERT model has a 92% accuracy in HTML analysis danger detection	Only works on english language and needs to expand to others Real-time database updates are needed to consistentl retrain and refit the models	MobileBERT and DistilBERT are better in detecting smishing attempts and can better adapt to evolving patterns used by attackers.
[1] Daniel Timko, Daniel Hernandez Castillo, and Muhammad Lutfur Rahman. 2024. A Quantitative Study of SMS Phishing Detection. https://doi.org/10.48550/arXiv.2311.06911	Identify factors that influence a users ability to differentiate between legitimate and false SMS messages Bridge current knowledge gaps on user perspective on smishing messages Inform of security strategies related to smishing	Participants were able to detect fake messages better 67.1% accuracy and relied on the sender information and URL to decide Those with good attention control and security behavioral intentions scored significantly higher in detection accuracy Age, income, occupation impacted interaction rates and accuracy.	Focus on user attention areas: users should focus on the sender information and URL to determine if the message is a scam or not Improved training: User training programs can help individuals recognizie phishing attempts Proactive defense: more data can help security experts design more effective warnings.	187 participants
[1] Iddi S. Mambina, Jema D. Ndibwile, and Kisangiri F. Michael. 2022. Classifying Swahili Smishing Attacks for Mobile Money Users: A Machine-Learning Approach. IEEE Access 10, (2022), 83061–83074. https://doi.org/10.1109/ACCESS.2022.3196464				
[1] Caner Balim and Efnan Sora Gunal. 2019. Automatic Detection of Smishing Attacks by Machine Learning Methods. In 2019 1st International Informatics and Software Engineering Conference (UBMYK), November 2019. 1–3. https://doi.org/10.1109/UBMYK48245.2019.8965429				
[1] Jae Woo Seo, Jong Sung Lee, Hyunwoo Kim, Joonghwan Lee, Seongwon Han, Jungil Cho, and Choong-Hoon Lee. 2024. On-Device Smishing Classifier Resistant to Text Evasion Attack. IEEE Access 12, (2024), 4762–4779. https://doi.org/10.1109/ACCESS.2024.3349577				